

Last login: Wed Sep 23 12:11:15 on ttys002
/var/folders/g6/2h7p_f2x2qbcf0_rwd7zs4y00000gn/T/orbstack-open-terminal_DBF0B79F.sh ; exit;
harsha@Harshas-MacBook-Air-3 ~ % /var/folders/g6/2h7p_f2x2qbcf0_rwd7zs4y00000gn/T/orbstack-open-terminal_DBF0B79F.sh ; exit;

OrbStack update available!
Run "orb update" to update.

Updates include improvements, features, and fixes.

```
harsha@Harsha:~$ nano memory_alloc.c
harsha@Harsha:~$ gcc -g -o memory_alloc memory_alloc.c
harsha@Harsha:~$ ./memory_alloc
malloc(): 1 2 3 4 5
calloc(): 0 0 0 0 0
realloc(): 1 2 3 4 5 6 7 8 9 10
Memory successfully freed.
harsha@Harsha:~$ valgrind --version
valgrind-3.22.0
harsha@Harsha:~$ valgrind --leak-check=full ./memory_alloc
==350== Memcheck, a memory error detector
==350== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==350== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==350== Command: ./memory_alloc
==350==
malloc(): 1 2 3 4 5
calloc(): 0 0 0 0 0
realloc(): 1 2 3 4 5 6 7 8 9 10
Memory successfully freed.
==350==
==350== HEAP SUMMARY:
==350==    in use at exit: 0 bytes in 0 blocks
==350==    total heap usage: 4 allocs, 4 frees, 1,104 bytes allocated
==350==
==350== All heap blocks were freed -- no leaks are possible
==350==
==350== For lists of detected and suppressed errors, rerun with: -s
==350== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
harsha@Harsha:~$ nano cow.c
harsha@Harsha:~$ gcc -o cow cow.c
harsha@Harsha:~$ ./cow
Parent PID: 357
Memory allocated: 3 MB

Parent process PID: 357
Child PID: 356

Child process PID: 356
Child initially shares memory pages with parent (Copy-on-Write).
Child modified the allocated memory.
data[0] = 0
Parent still has data[0] = 0
harsha@Harsha:~$ pmap -x 400
harsha@Harsha:~$ pmap -x 401
harsha@Harsha:~$
```